

KANGWA MUKUKA

Website: kangwa-mukuka.web.app | Email: kangwa1mukuka@cs.unza.zm | +260 964475128 | linkedin.com/in/kmukuka

A Student of Computer Science passionate about building useful solutions that drive social impact in resource-constrained settings. At DataLab Research Group, I translate real-world operational challenges into AI-supported workflows and work on projects at the cusp of Machine Learning and Artificial Intelligence for Healthcare in Zambia and Africa.

EDUCATION

Bachelor of Science - BSc., Computer Science | Major: Computer Systems Engineering Feb 2022 - May 2026

University of Zambia (UNZA) | Lusaka, ZM

Prj: *Enhancing Radiology Workflows: Semi-automated Cervical Cancer Reporting at the Cancer Diseases Hospital (Zambia)*

Supervisors: [Dr. Lighton Phiri](#), [Dr. Mayumbo Nyirenda](#)

HONORS AND AWARDS

Undergraduate Student Innovator of the Year Award UNZA (2025), [African Network for Artificial Intelligence in Biomedical Imaging \(AFRICAI\) Mentorship Award \(2025\)](#), 3rd place: IEEE IAS IPCSD Engineering Design Contest Africa (2024), 3rd place: Zambia@60 National ICT Hackathon (2024), 1st place: UNZA Student Innovation Challenge (2025).

RELEVANT COURSEWORK

Fundamentals of Artificial Intelligence | Data Mining & Warehousing | IT Project Management | Computer Programming

SKILLS

Technical Skills: Python, JavaScript, Java, C++, Flask, React, CSS, Bootstrap, MySQL, PyTorch.

Tools: Git, GitHub, Firebase, Google Colab, Jupyter Notebook.

Soft Skills: Research Methods, Workflow Analysis, Requirements Gathering, Teamwork, Stakeholder Engagement.

WORK AND LEADERSHIP EXPERIENCE

Research Assistant - Enterprise Medical Imaging | DataLab Research Group

Sep 2025 - Present

University of Zambia | Lusaka, ZM

- Contributing to the research and integration of Vision Language Models to semi-automate cervical cancer report generation at the Cancer Diseases Hospital, with the aim of reducing the time to 1/3, addressing the 3,600+ annual cervical cancer cases in Zambia. I was selected as one of 10 mentees across Africa by [AFRICAI](#).
- Secured key collaboration with [Dawa Health](#), working towards a unified screening tool for the early detection of cervical cancer among women in Zambia, Zimbabwe, and Rwanda, where 8,000+ combined cases occur annually.

Chairperson

Dec 2024 - Nov 2025

Institute of Electrical and Electronics Engineers (IEEE) UNZA Student Branch | Lusaka, ZM

- Led an executive team of 11 students and an association of 560+ student members. Coordinated industrial tours and participation in continental contests, emerging 3rd across Africa in one of them. Grew participation by 20%.

Technical Lead

Sep 2023 - Jun 2024

Google Developer Student Club (GDSC) UNZA Student Chapter | Lusaka, ZM

- Engaged 55+ students in technical activities by conducting information sessions and hosting fireside chats with inspiring Software Engineering industry leaders. I actively recruited GDSC members during 1st year orientation.

FEATURED PROJECTS

Mota50 - Fleet Management System | World Vision Zambia (WVZ) Innovation Challenge Nov 2025 - Present

- Built to streamline fleet management operations at WVZ, innovating for development challenges in Zambia.

Enterprise Medical Imaging in Zambia (EMI x Z) | AI Supported Workflows [emi.org.zm] Mar 2025 - Present

- This project seeks to investigate the challenges resulting from the shortage of radiologists and explores the feasibility of implementing efficient and effective medical imaging workflows using enterprise imaging techniques.

ReviZa | Transforming the Future of Education in Zambia [reviza-31466.web.app] Aug 2024 - Mar 2025

- Spearheaded the product/strategy of this platform, facilitating knowledge-sharing of academic resources. Gained 100+ active users in the 1st month. Emerged 3rd of 89 teams in a National Hackathon ([Ministry of Education](#)).
- Translated student users' needs into core features for ease-of-use and searchability/discoverability of resources.

PUBLICATIONS | PEER-REVIEWED CONFERENCE PAPERS

Design and Prototyping of a Remote Livestock Monitoring Device for Low-Resource Settings

- M. M. Makumba, J. C. Moya, D. Banda, and K. E. Mukuka, IEEE International Humanitarian Technology Conference (IHTC), Edmonton, AB, Canada, 2025, pp. 44-48, doi: [10.1109/IHTC65087.2025.11216306](https://doi.org/10.1109/IHTC65087.2025.11216306).

Design of a Low-Cost Smart Solar-Powered Irrigation System

- K. M. Makasa, and K. E. Mukuka, IEEE PES/IAS PowerAfrica Conference (PAC), Johannesburg, South Africa, 2024, pp. 1-5, doi: [10.1109/PowerAfrica61624.2024.10759498](https://doi.org/10.1109/PowerAfrica61624.2024.10759498).